

THREAD STATES & TRANSITIONS

Instructor:



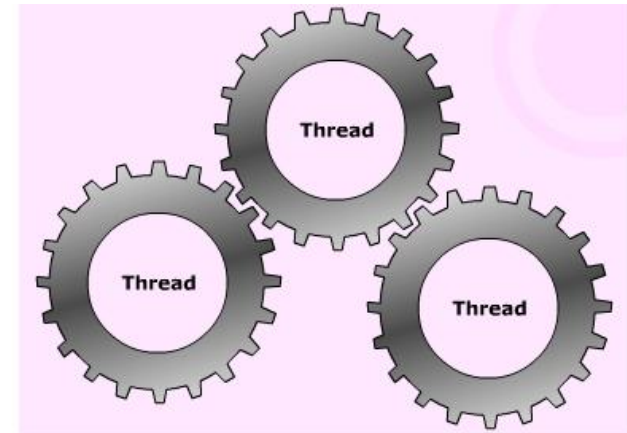
- ◇ **Introduction to Thread**
- ◇ **Creating Threads**
 - ✓ **Thread class**
 - ✓ **Runnable interface**
- ◇ **Thread States and Transitions**
- ◇ **Managing Thread**



Section 1

INTRODUCTION TO THREAD

- A **process** is a program that is executing.
 - Each process has its own run-time resources, such as their own data, variables and memory space.
 - Each process executes **several tasks** at a time and each task is carried out by separate **thread**.
- A **thread** is a **path of code execution** through a program, and each thread has its own *local variables*, *program counter*, and *lifetime*.



Process and Thread

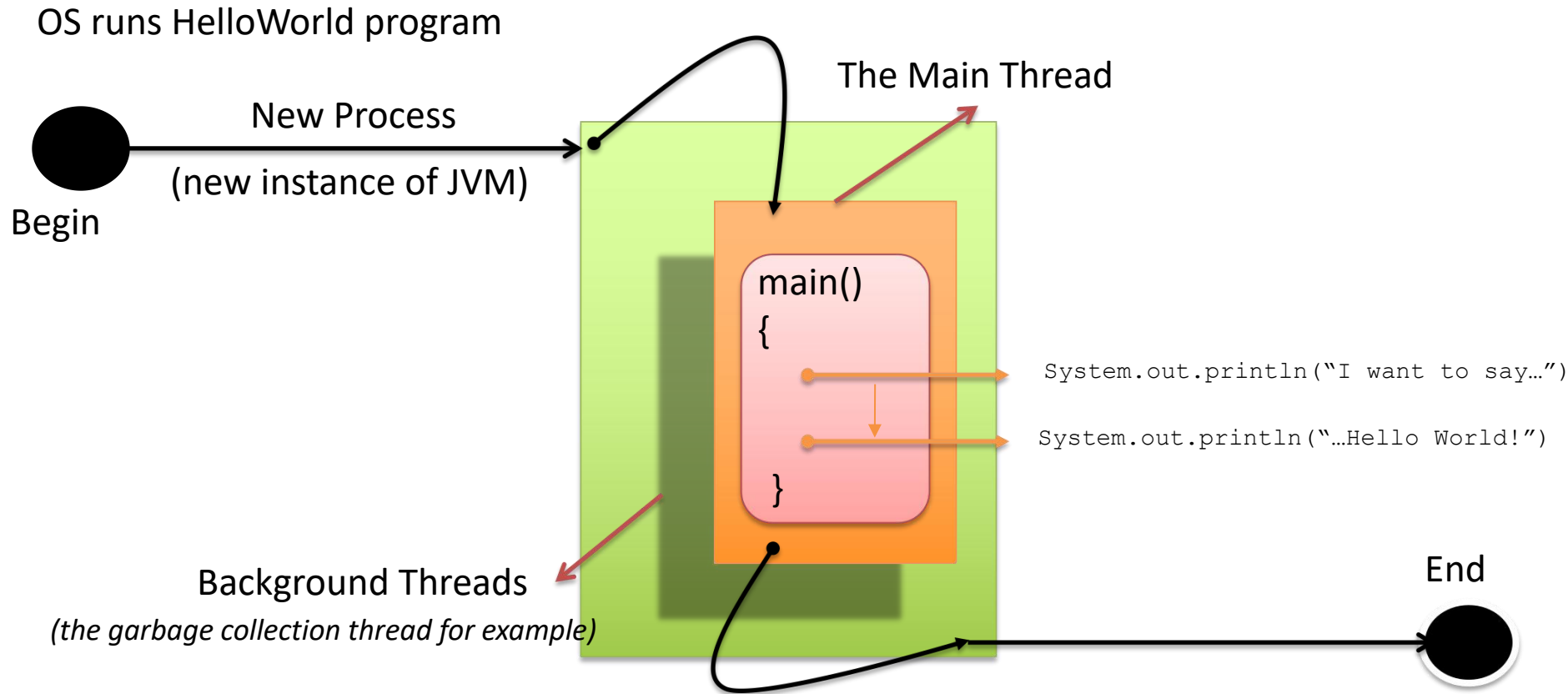
- **A thread** has its own complete set of **basic run-time resources** to run it independently.
- **A thread is the smallest unit of executable code** in an application that performs a particular job or task.
- **Several threads can be executed at a time**, facilitating execution of several tasks of a single application simultaneously.
- **Threads are independent**, if there occurs exception in one thread, it doesn't affect other threads. It shares a common memory area.

Let's try HelloWorld again!

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("I want to say...");  
        System.out.println("...Hello World!");  
    }  
}
```

```
C:\java HelloWorld ↵  
I want to say...  
...Hello World!
```

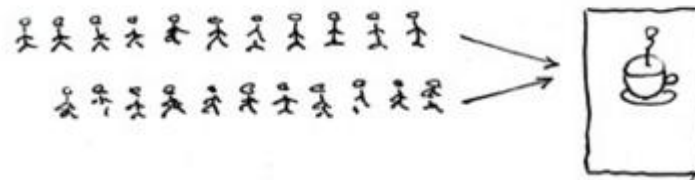
What's happening?



Even a simple HelloWorld program is running in a multithreaded environment

- Concurrent programming is a process of **running several tasks at a time**.

Concurrent = Two Queues One Coffee Machine



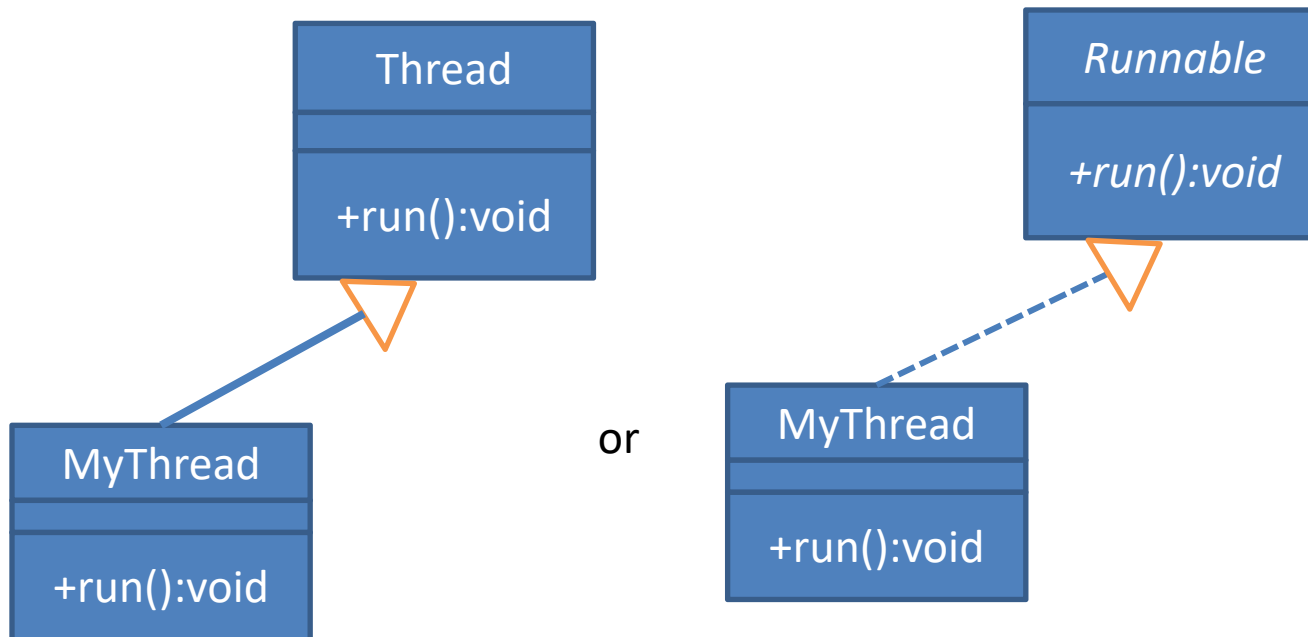
- In Java**, it is possible to execute simultaneously^[đồng thời] a invoked function without waiting for the invoked function to terminate.
- The invoked function runs independently and concurrently with the invoking program, and can share variables, data and so on with it.

Section 2

CREATING THREADS

How to create a Thread?

- Which way you can create a thread?
 - ✓ Inherits the **Thread** class
 - ✓ Implements the **Runnable** interface



Subclassing the Thread class

- **Step 1:** Subclassing Thread class
- **Step 2:** Override run() method

```
// Extending Thread class
class MyThread extends Thread {
    public void run()    { // Overriding the Run()
        // implementation the logic
    }
}
```

- **Step 3:** Create a thread object and start the thread

```
public static void main(String args[]) {
    //Creating thread object
    MyThread t = new MyThread();
    t.start(); //Starting a thread
}
```

Implementing *Runnable* interface

- ❖ **Step 1:** Implement the *Runnable* interface
- ❖ **Step 2:** Implement *run()* method

```
// Declaring a class that implements Runnable interface
class MyRunnable implements Runnable {
    public void run()    { // Overriding the Run()
        // implementation the logic
    }
}
```

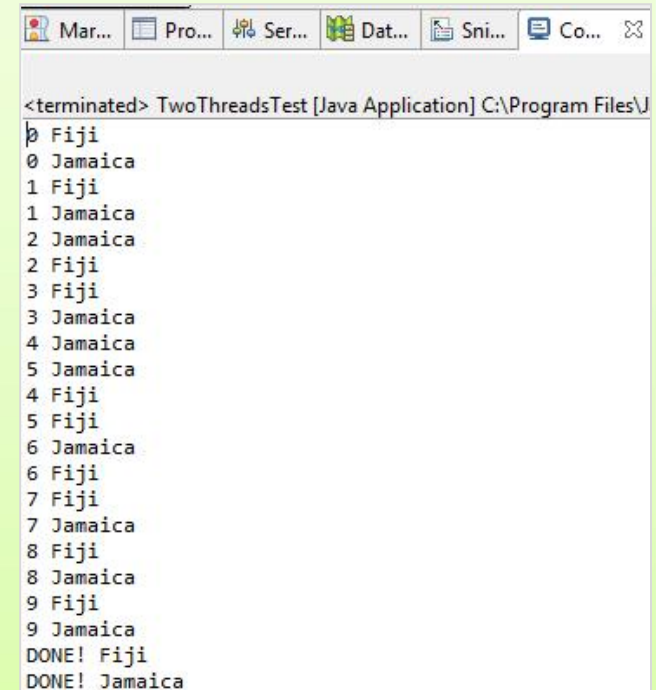
- ❖ **Step 3:** Create a thread object and start the thread

```
public static void main(String args[]) {
    Runnable r = new MyRunnable();
    Thread thObj=new Thread(r);
    thObj.start(); //Starting a thread
}
```

Create a Thread: Demo

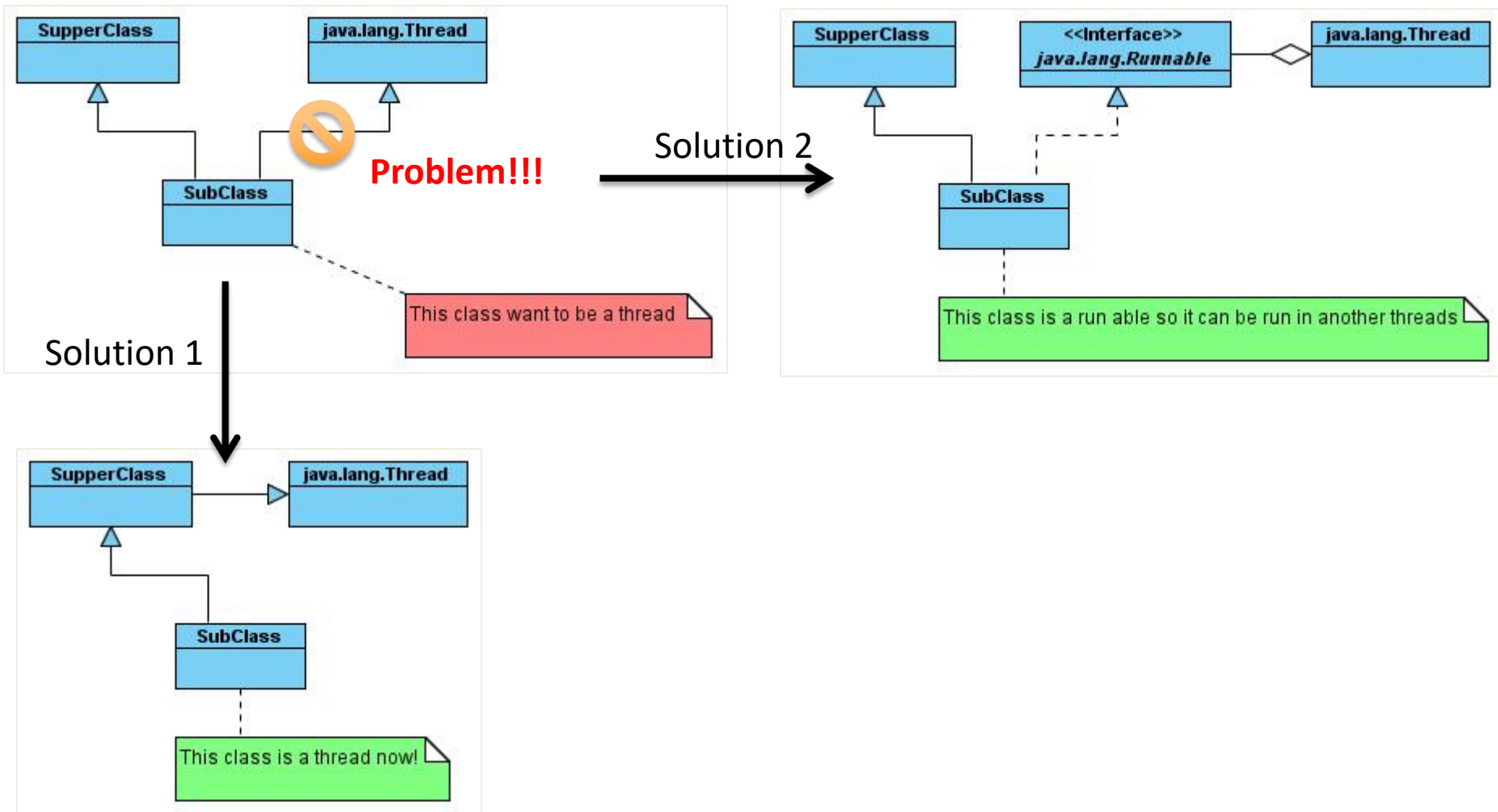
Code snippet

```
public class TwoThreadsTest {  
    public static void main(String[] args) {  
        new SimpleThread("Jamaica").start();  
        new SimpleThread("Fiji").start();  
    }  
}  
  
class SimpleThread extends Thread {  
    public SimpleThread(String str) {  
        super(str);  
    }  
  
    public void run() {  
        for (int i = 0; i < 10; i++) {  
            System.out.println(i + " " + getName());  
            try {  
                sleep((int) (Math.random() * 1000));  
            } catch (InterruptedException e) {  
            }  
        }  
        System.out.println("DONE! " + getName());  
    }  
}
```



```
<terminated> TwoThreadsTest [Java Application] C:\Program Files\J  
0 Fiji  
0 Jamaica  
1 Fiji  
1 Jamaica  
2 Jamaica  
2 Fiji  
3 Fiji  
3 Jamaica  
4 Jamaica  
5 Jamaica  
4 Fiji  
5 Fiji  
6 Jamaica  
6 Fiji  
7 Fiji  
7 Jamaica  
8 Fiji  
8 Jamaica  
9 Fiji  
9 Jamaica  
DONE! Fiji  
DONE! Jamaica
```

Implementing Runnable

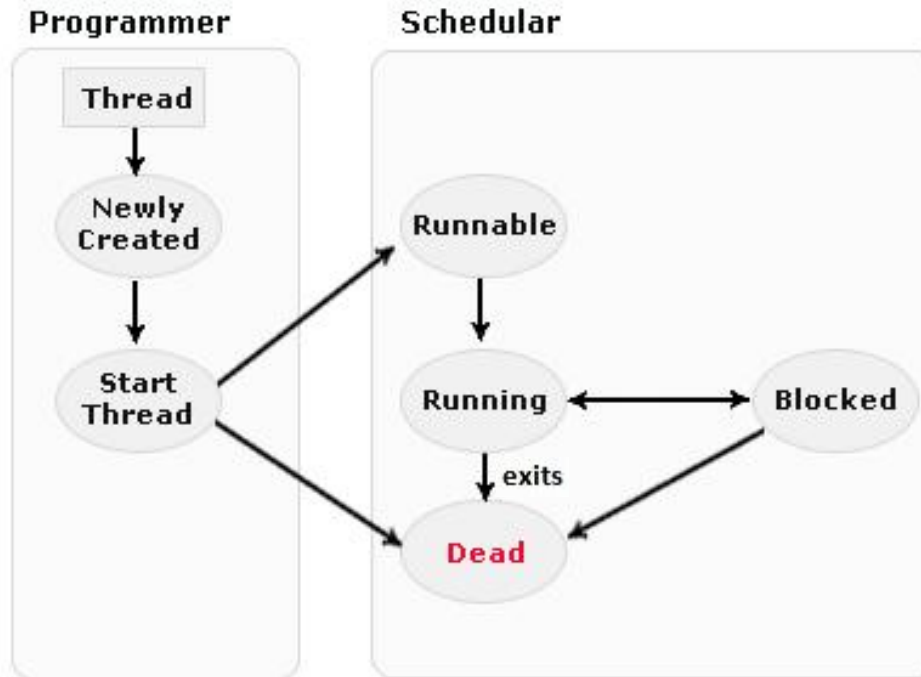


Section 3

THREAD STATES AND TRANSITIONS

Thread States

- Difference state of a thread are:



1. New state
2. Runnable (Ready-to-run) state
3. Running state
4. Blocked
5. Dead state

- **Thread** class provide ***constructors*** and ***methods*** to create and perform operations on a thread. Thread class extends Object class and implements Runnable interface.
- Commonly used Constructors:
 - ✓ Thread()
 - ✓ Thread(String name)
 - ✓ Thread(Runnable r)
 - ✓ Thread(Runnable r,String name)

Methods of the Thread class

Method	Return Type	Description
static currentThread()	Thread	Returns an object reference to the thread in which it is invoked.
getName()	String	Retrieve the name of the thread object or instance.
start()	void	starts the execution of the thread. JVM calls the run() method on the thread.
run()	void	This method is the entry point to execute thread, like the main method for applications.
static sleep()	void	Suspends a thread for a specified amount of time (in milliseconds).
isAlive()	boolean	This method is used to determine the thread is running or not.
static activeCount()	int	This method returns the number of active threads in a particular thread group and all its subgroups.
interrupt()	void	The method interrupt the threads on which it is invoked.
static yield()	void	By invoking this method the current thread pause its execution temporarily and allow other threads to execute.
join()	void	This method and join(long millisec) Throws InterruptedException. These two methods are invoked on a thread instance, the currently running thread will block until the Thread instance has finished executing.

- **start() method** of Thread class is used to start a newly created thread. It performs following tasks.
 - ✓ A new thread starts (with new **callstack**).
 - ✓ The thread moves from New state to the Runnable state.
 - ✓ When the thread gets a chance to execute, its target run() method will run.

Code snippet

```
public class MyThread extends Thread {  
    @Override  
    public void run() {  
        System.out.println("Thread is running...");  
    }  
  
    public static void main(String[] args) {  
        MyThread myThread = new MyThread();  
        myThread.start();  
    }  
}
```

Starting a Thread (2/4)

- Be sure that, you **start a Thread, not a Runnable**. You call start() on a Thread instance, not on a Runnable instance.

Code snippet

```
class FooRunnable implements Runnable {  
    public void run() {  
        for (int x = 1; x < 6; x++) {  
            System.out.println("Runnable running");  
        }  
    }  
}
```

```
public class TestThreads {  
    public static void main(String[] args) {  
        FooRunnable r = new FooRunnable();  
        Thread t = new Thread(r);  
        t.start();  
    }  
}
```

```
Runnable running  
Runnable running  
Runnable running  
Runnable running  
Runnable running
```

- If you see code that calls the `run()` method on a `Runnable` (or even on a `Thread` instance), **that's perfectly legal**.
 - ✓ But it **doesn't** mean the `run()` method will run in a **separate thread**!
 - ✓ Calling a `run()` method directly just means you're invoking a method from whatever **thread is currently executing**, and the **`run()` method goes onto the current call stack rather than at the beginning of a new call stack**.
- The following code does not start a new thread of execution:

```
Thread t = new Thread();  
t.run(); // Legal, but does not start a new thread
```

Starting a Thread (4/4)

Watch

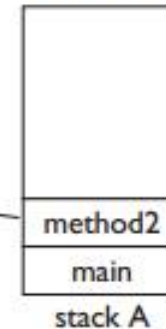
FIGURE 9-1

Starting a thread

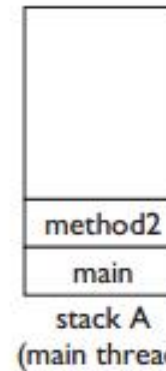
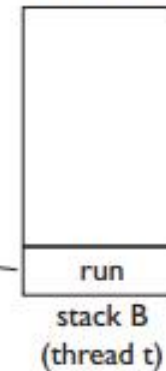
```
public static void main(String [] args) {  
    // running  
    // some code  
    // in main()  
    method2();  
    // running  
    // more code  
}  
  
static void method2() {  
    Runnable r = new MyRunnable();  
    Thread t = new Thread(r);  
    t.start();  
    // do more stuff  
}
```



1) main() begins



2) main() invokes method2()



3) method2() starts a new thread

sleep() Method

- **Thread.sleep()** suspends^[ngừng/hoãn] the execution of the current thread for a specified period of time.
 - ✓ It makes the **processor time available** to the **other threads** of an application or **other applications**.
 - ✓ It stops the execution if the active thread for the time specified in milliseconds or nanoseconds.
- It raises **InterruptedException** when it is interrupted using the interrupt () method.

Code snippet

```
public void run()
{
    for(int x=1; x<4; x++) {
        System.out.println("nameRunnable is running by" +
        /** getName() method give the name of the
        * currently running thread.*/
        Thread.currentThread().getName());
        try{
            Thread.sleep(1000);
            /** sleep() method will force the running thread to
            * wait for 1000 milli seconds.*/
        }catch (InterruptedException ex) { }
    }
}
```

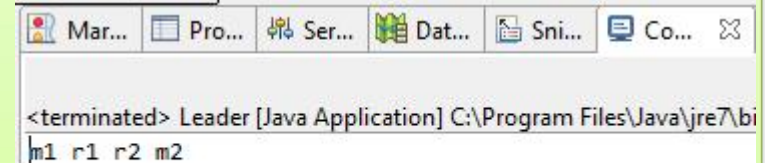
An example of using sleep() method

join() Method (1/2)

- The join() method is used to hold the execution of currently running thread until the specified thread is dead (finished execution).

Code snippet

```
public class Leader implements Runnable {
    public static void main(String[] args) {
        Thread t = new Thread(new Leader());
        t.start();
        System.out.print("m1 ");
        try {
            t.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
        System.out.print("m2 ");
    }
    public void run() {
        System.out.print("r1 ");
        System.out.print("r2 ");
    }
}
```



<terminated> Leader [Java Application] C:\Program Files\Java\jre7\bin
m1 r1 r2 m2

- If you have a **thread B** that can't do its work until another **thread A** has completed its work, then you want thread B to "join" thread A. This means that thread B will not become runnable until A has finished (and entered the dead state).

```
Thread t = new Thread();
```

```
t.start();
```

```
t.join();
```

“the code `t.join()` means "Join me (the current thread) to the end of `t`, so that `t` must finish before I (the current thread) can run again”

- Exer: Viết 1 thread để gán giá trị ngẫu nhiên cho một biến, running thread và lấy giá trị của biến.

- By invoking this method the current thread pause its execution temporarily and allow other threads to execute.
 - ✓ Yield is a **Static** method and **Native** too.
 - ✓ Yield tells the **currently executing thread** to give a chance to the threads that have **equal priority in the Thread Pool**.
 - ✓ There is **no guarantee** that Yield will make the currently executing thread to runnable state **immediately**.
 - ✓ It can only make a thread **from Running State to Runnable State**, not in wait or blocked state.

yield() Method (2/2)

Example

Code snippet

```
public class YieldExample {  
    public static void main(String[] args) {  
        Thread producer = new Producer();  
        Thread consumer = new Consumer();  
  
        producer.setPriority(Thread.MIN_PRIORITY); // Min Priority  
        consumer.setPriority(Thread.MAX_PRIORITY); // Max Priority  
  
        producer.start();  
        consumer.start();  
    }  
}
```

```
class Producer extends Thread {  
    public void run() {  
        for (int i = 0; i < 5; i++) {  
            System.out.println("I am Producer :  
                Produced Item " + i);  
            Thread.yield();  
        }  
    }  
}
```

```
class Consumer extends Thread {  
    public void run() {  
        for (int i = 0; i < 5; i++) {  
            System.out.println("I am Consumer :  
                Consumed Item " + i);  
            Thread.yield();  
        }  
    }  
}
```

isAlive() Method

- The **main thread** must be the last thread to finish.
- A thread is considered to be alive when it is running. We can check it by use isAlive() method.
- This method helps one thread to check state of another **to do something** or **use join() method**.
- **Syntax:**

```
System.out.println("The thread is alive: "  
                    +thread.isAlive());
```

isAlive() Method (2/2)

Code snippet

```
public class IsAliveTest {  
    public static void main(String args[]) {  
        SimpleThread thread1 = new SimpleThread("One");  
        SimpleThread thread2 = new SimpleThread("Two");  
        System.out.println("Thread One is alive: "+ thread1.isAlive());  
        System.out.println("Thread Two is alive: "+ thread2.isAlive());  
        // wait for threads to finish  
        try {  
            System.out.println("Waiting for threads to finish.");  
            thread1.join();  
            thread2.join();  
        } catch (InterruptedException e) {  
            System.out.println("Main thread Interrupted");  
        }  
        System.out.println("Thread One is alive: "+ thread1.isAlive());  
        System.out.println("Thread Two is alive: "+ thread2.isAlive());  
        System.out.println("Main thread exiting.");  
    }  
}
```

Section 4

MANAGING THREAD

- **Thread priority** helps the thread scheduler to decide which thread to run.
- Priority also helps the OS to decide the amount of **resource** that has to be allocated to each thread.
- Thread priority:
 - ✓ Thread.MAX_PRIORITY : constant value of 10
 - ✓ Thread.NORM_PRIORITY : constant value of 5, default
 - ✓ Thread.MIN_PRIORITY : constant value of 1
- Important Methods:
 - ✓ setPriority()
 - ✓ getPriority()

Code snippet

```
public class ThreadDefaultPriority extends Thread {  
    public void run() {  
        System.out.println(getName() + ":" +  
            Thread.currentThread().getPriority());  
    }  
  
    public static void main(String[] args)  
        throws InterruptedException {  
        ThreadDefaultPriority t1 = new ThreadDefaultPriority();  
        ThreadDefaultPriority t2 = new ThreadDefaultPriority();  
  
        t1.start();  
        t2.start();  
    }  
}
```

```
Thread-1:5  
Thread-0:5
```

Code snippet

```
public class ThreadPriority extends Thread {  
    public void run() {  
        String tName = Thread.currentThread().getName();  
        Integer tPrio = Thread.currentThread().getPriority();  
        System.out.println(tName + " has priority " + tPrio);  
    }  
  
    public static void main(String[] args) throws InterruptedException {  
        ThreadPriority t0 = new ThreadPriority();  
        ThreadPriority t1 = new ThreadPriority();  
        ThreadPriority t2 = new ThreadPriority();  
  
        t1.setPriority(Thread.MAX_PRIORITY);  
        t0.setPriority(Thread.MIN_PRIORITY);  
        t2.setPriority(Thread.NORM_PRIORITY);  
  
        t0.start();  
        t1.start();  
        t2.start();  
    }  
}
```

```
Thread-1 has priority 10  
Thread-0 has priority 1  
Thread-2 has priority 5
```

- If a thread enters the **runnable** state, and it has a higher priority than any of the threads in the pool and a higher priority than the currently running thread:
 - ✓ the **lower-priority running thread** usually will be bumped **back to runnable** and the **highest-priority thread will be chosen to run**.
- At any given time the currently running thread usually will not have a priority that is lower than any of the threads in the pool.
 - ✓ In most cases, the running thread will be of equal or greater priority than the highest priority threads in the pool.
- **Don't rely on thread priorities when designing your multithreaded application.**
 - ✓ **Use thread priorities as a way to improve the efficiency of your program.**

- In Java **Daemon threads** are service providers for normal running threads inside the same process. Java Daemon threads does their execution as helper to complete the current task.
- The characteristics of the daemon threads are:
 - ✓ They work in the background providing service to other threads.
 - ✓ They are fully dependent on the user threads.
 - ✓ JVM stops once a thread dies and only daemon thread is alive.
- To specify a thread is deamon or not, **setDaemon()** method is used

Code snippet

```
public class Main {  
    public static void main(String[] args) {  
        BackgroundService service = new BackgroundService();  
        try {  
            Thread.sleep(10000);  
        } catch (InterruptedException ex) {  
            System.out.println(ex);  
        }  
    }  
}
```

```
public class BackgroundService extends Thread {  
    private int count;  
    public BackgroundService() {  
        setDaemon(true);  
        start();  
    }  
  
    @Override  
    public void run() {  
        while (true) {  
            System.out.println("#" + count++);  
            // Causes the currently executing thread object to temporarily pause  
            // and allow other threads to execute.  
            yield();  
        }  
    }  
}
```

- The tasks performed by the Daemon threads are:
 - ✓ *Daemon threads are service providers for other threads running in the same process,*
 - ✓ *Daemon threads are designed as low-level background threads that perform some tasks such as mouse events for Java program.*

- **Exer2:** Tạo một Daemon Thread để thêm vào List một danh sách số trong khi User Thread đang thực hiện một tác vụ khác.

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Thank you

